MULTI·LLL 2026 · WORKSHOP TALK

Clean Evaluations on Contaminated *Visual* Language Models

Can we still trust VLM leaderboard numbers?

Hongyuan Lu
FaceMind · Equal contribution

Shujie Miao
National University of Singapore · Equal contribution

Wai Lam
The Chinese University of Hong Kong

The *score* wasn't earned.

Data contamination — when benchmark images leak into pretraining, the metric on the leaderboard stops measuring the model's true capability.

INTERNVL · EPOCH 5 · BLEU INFLATION

+0.0452

0.1047 clean → **0.1499** contaminated · +43% relative

01 Training set overlaps the test set.

02 Scores look better than the model's real ability.

03 The leaderboard number stops being what it claims.

Rich on text, *thin on vision*.

	TEXT LLMS *three categories of work, mature and tooled*	VISUAL LMS *almost unaddressed*
Detection	**CDD · TED · Min-K%++** Look at output distributions to flag contamination.	*— unaddressed*
Prevention	**DyVal · withhold test data** Build evaluation data after model release; dynamic test generation.	*— unaddressed*
Clean eval	**Clean-Eval** Text rephrasing for surface-level different prompts.	**This work** Image-side probe via augmentation, BGR as a robust choice.

Audit the *image*. Keep the *question*.

At evaluation time we transform only the visual input. The text query x stays fixed.

ORIGINAL

(x, z)

text query + raw image

what the model already saw

t
IMAGE ONLY →

TRANSFORMED

(x, z')

same x · image underwent t

flips · rotations · channel swap · …

→

CALIBRATED SCORE

$P(y \mid x, z')$

model M processes (x, z′) — the output y is the calibrated evaluation

a transformed image cannot be a memorised image.

$$(x, z) \rightarrow (x, z' = t(z)) \cdot t \in T, \quad x \text{ unchanged}$$

A contaminated model can still answer the leaked image. It cannot as easily answer its transformed self.

Built *after* the models it tests.

Collection dates 20–25 ***June*** 2024 — after every model under study had already been released.

STEP 01

1,000

in-game screenshots

Gamersky guide site

▶

STEP 02

1–3

QA pairs per image

GPT-4o multimodal drafts

▶

STEP 03

manual

review & correction

every pair, by a human

▶

FINAL

2,561

dialogue rounds

images + QA, paired

SPLIT



90% train

10% test

Fine-tuned via LoRA, used both to train M1 and to swap into M2.

COLLECTION WINDOW

20 – 25 Jun 2024

After bunny-4B & internvl2.0-2B had shipped.
Contamination avoided by construction.

What the *data* looks like.



Q: What is the central element in the background of the image?

A: Statue.

Q: How many columns are visible in the background of the image?

A: Four.

Q: What is the direction of the sunlight in the image?

A: Diagonal.



Q: What type of setting is shown in the image?

A: Garden.

Q: What is the main object being held in the image?

A: Gun.

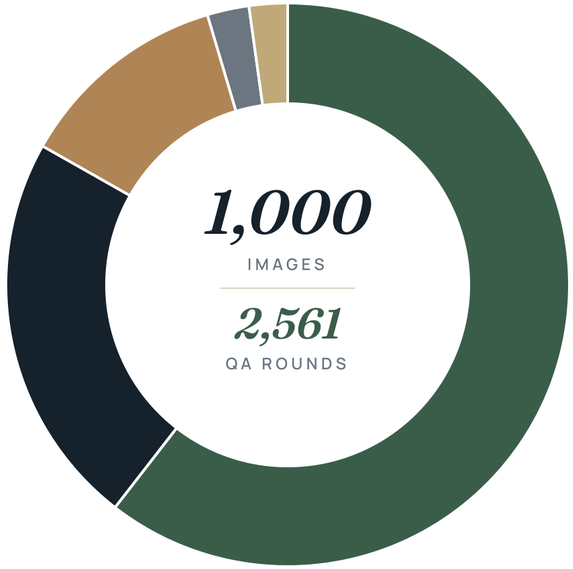
Q: Are there any flowers visible in the image?

A: Yes.

Figure 1 · Two illustrative image-to-QA pairs. Objects, scene text, and visual context all carry meaning.

Five genres, one *dominant* title.

1,000 in-game screenshots spanning open-world RPG, anime titles, shooters and more.



ELDEN RING

605

of 1,000 images — 60.5% of the corpus

Anime Games
Genshin Impact · Honkai: Star Rail
227
images · 22.7%

Shooting Games
GTA V · Valorant · Delta Force
122
images · 12.2%

Other RPG
Dungeon & Fighter
24 2.4%

Others
Palworld · League of Legends
22 2.2%

Same counts as the paper's Figure 2 — redrawn as a ring in this deck's palette.

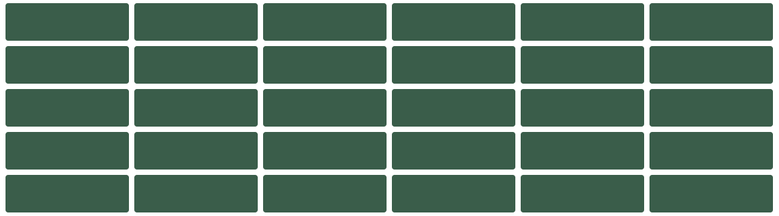
Two models. *Same* size, swapped contents.

Both trained for **5 and 10 epochs**; evaluated on the same held-out 10%.

MODEL M1

Clean baseline.

Trained on the training split only.
Test images **never seen** during training.



Training data (900 samples)

Held-out test set (100) — never enters training

MODEL M2

Contaminated twin.

Training kept at 900 samples, but a subset is **replaced** with test images before training.



Training data

Swapped-in test

Total training size is unchanged. Same held-out test set as M1.

Two VLMs, shared *protocol*.

Trained identically up to the contamination swap; evaluated on the held-out 10%.

MODEL 01

bunny-4B

Backbone	phi-2
LoRA	`rank 128 · alpha 256`
Learning rate	`5e-4`

MODEL 02

internvl2.0-2B

Backbone	frozen + LoRA on LM
LoRA	`rank 16 (LM head)`
Learning rate	`4e-5`

TRAINING & EVALUATION

Effective batch	Precision	Schedule	Epochs
16	`bf16 + ckpt`	`cosine · warmup`	`5 · 10`

HARDWARE

2 × RTX 4090

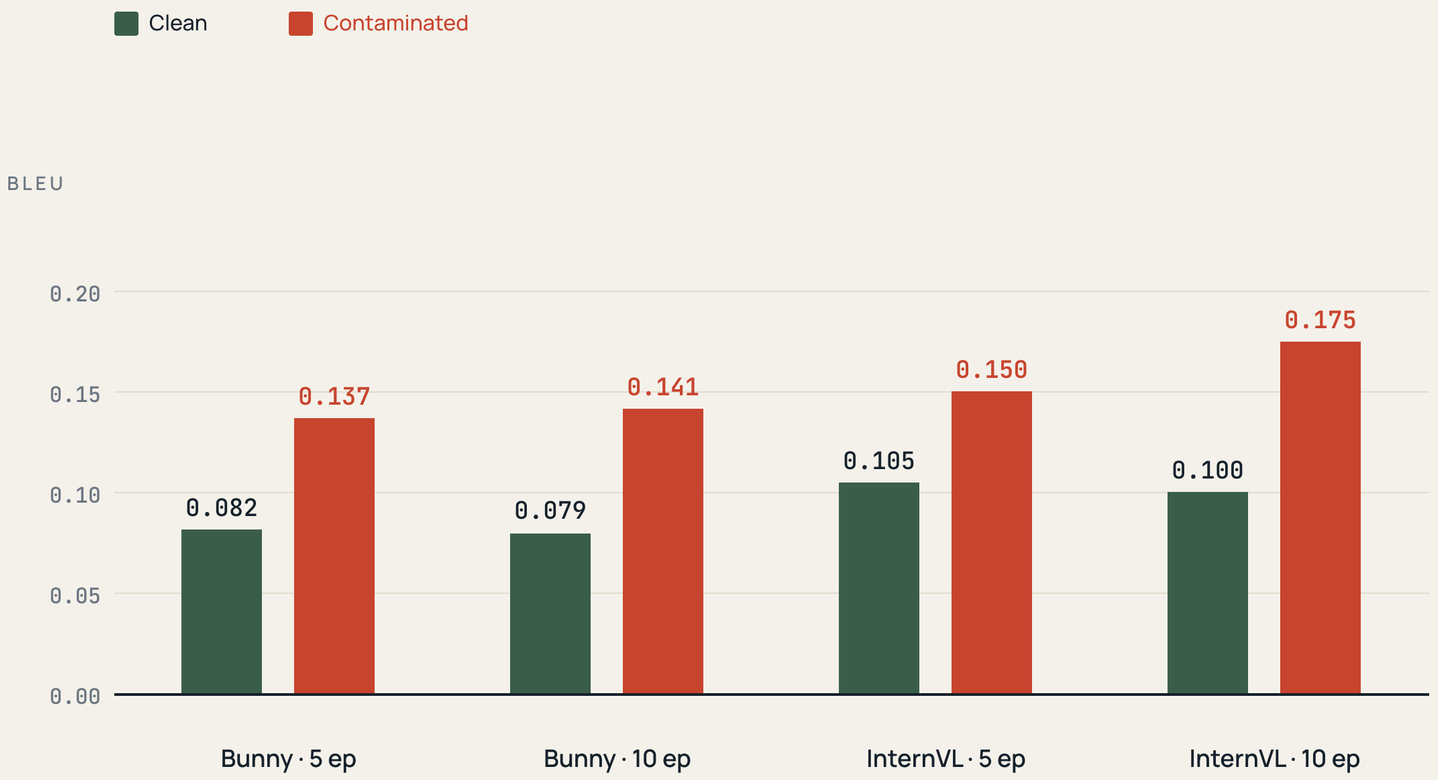
24 GB each · bf16 mixed precision

METRICS

BLEU	*ROUGE-1*	*ROUGE-2*
Papineni et al., 2002	unigram overlap, Lin 2004	bigram overlap, Lin 2004

Contamination *pays off* — everywhere it tries.

BLEU on the test split, every model–epoch pair, clean vs contaminated.



INFLATION GAP

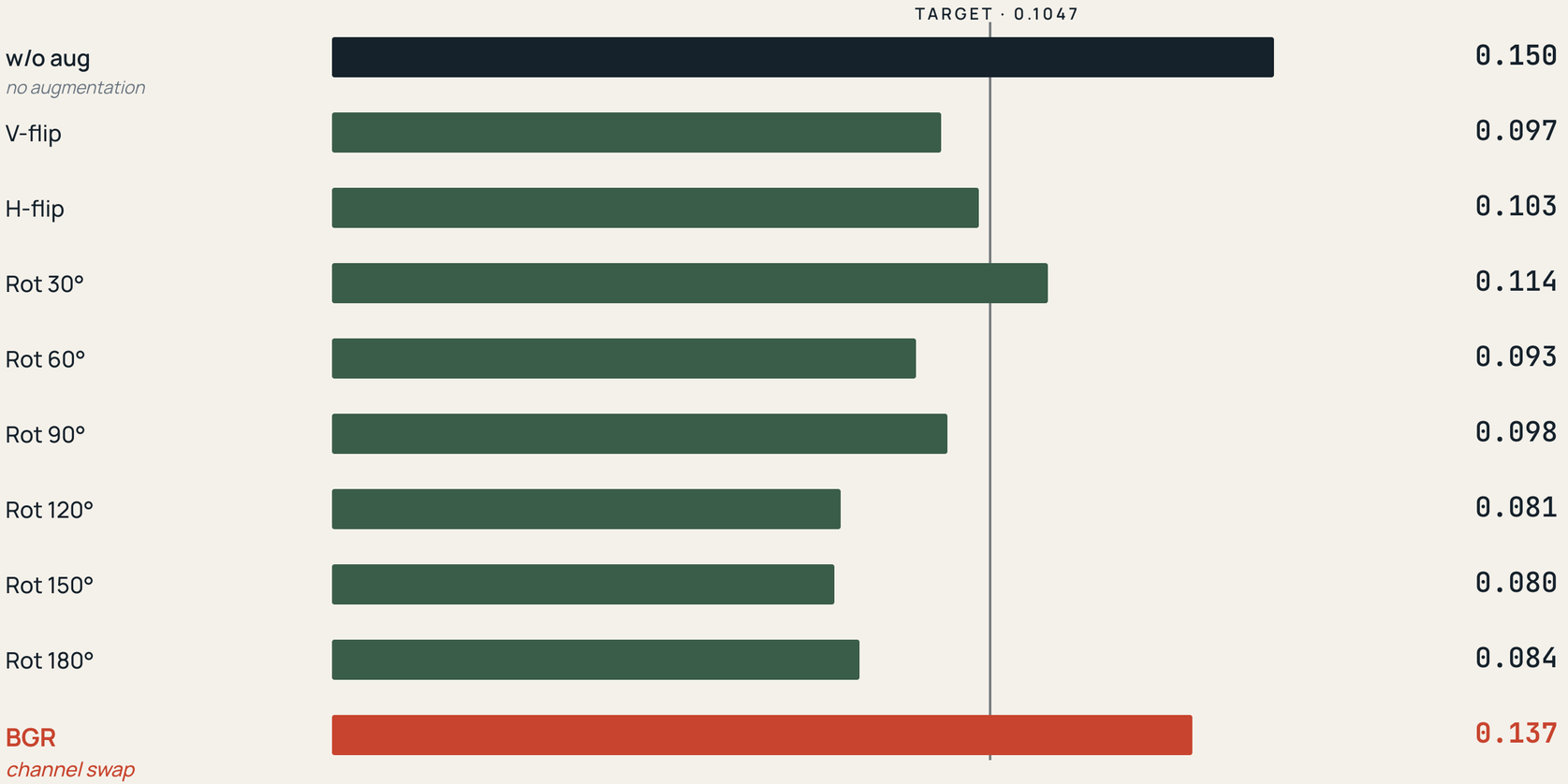
Every metric, every row.

GROUP	Δ BLEU	Δ R-1 / R-2
Bunny E5	+0.055	+0.216 +0.079
Bunny E10	+0.062	+0.226 +0.083
InternVL E5	+0.045	+0.162 +0.057
InternVL E10	+0.075	+0.286 +0.103

Bigger jump at longer training — more memorisation, more inflation.

Transforms claw inflated scores *back*.

InternVL · Epoch 5 · BLEU on the contaminated model, ten test-time augmentations.



Severity helps — but every augmentation already pulls the contaminated model below its own inflated peak.

But flips and rotations are *a trap*.

They look like a clean probe. They are also standard training augmentations.
A malicious trainer can simply add them to training.

STEP 01

The probe looked fair.

Flips and rotations pulled contaminated scores down — back toward the clean ones.

→

STEP 02

Trainer absorbs it.

Add the same augmentation to training data. The model learns the augmented world.

→

STEP 03

Scores re-inflate.

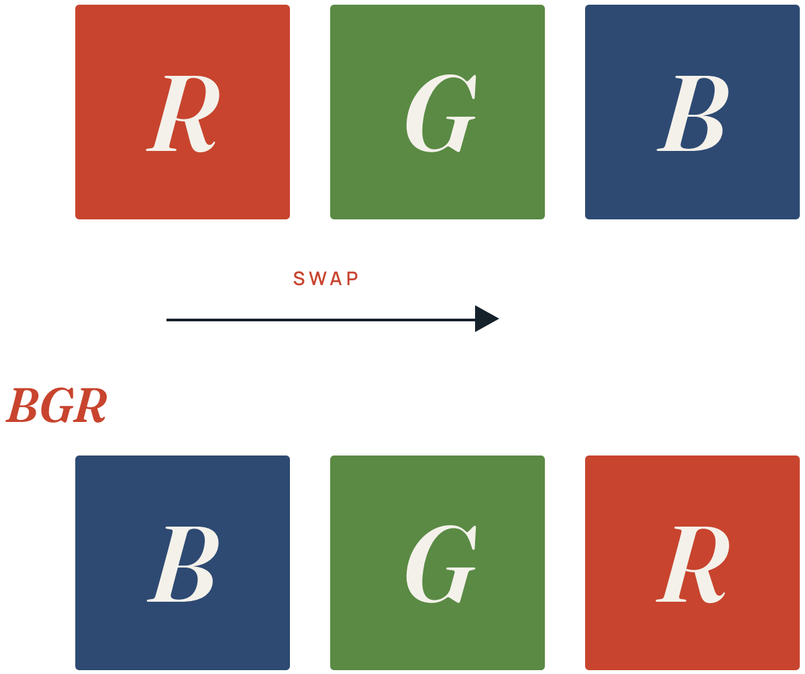
The probe is now part of the training routine. The leaderboard goes back up.

So which augmentation can a trainer not absorb?

BGR — swap the *channel order*.

Human eyes can still parse the image; model pixel-statistic shortcuts break.

CHANNEL REORDER



EFFECT AT EVALUATION

Contaminated InternVL · Epoch 5 · BLEU

Contaminated, no aug

0.1499

Under BGR

0.1368

CLEAN TARGET

0.1047

what the clean run produces

Closest of all ten. Not yet clean.

Still `+0.0321` above target — honest about that gap.

A semantic stress: human eyes barely notice; the model's pixel-statistic shortcuts do.

A probe trainers *cannot* absorb.

InternVL · Epoch 5 · what happens when BGR-augmented examples are mixed into training.

METRIC	ORIGINAL	MIXED BGR	Δ
BLEU	0.1047	0.1041	-0.0006 *essentially no gain*
ROUGE-1	0.6250	0.6082	-0.0168 *a small drop*
ROUGE-2	0.0670	0.0670	0.0000 *exactly equal*

Train on BGR or don't — performance does not improve.

A probe trainers have **no incentive** to bake in.

Three contributions, *one* honest limit.

01

A new visual clean evaluation benchmark.

1,000 images · 2,561 dialogue rounds · collected after model release.

02

Image augmentation as a probe.

Transform only the image; keep the question. Pulls inflated scores back toward real capability.

03

BGR: a clean probe that trainers cannot absorb.

Closest to clean — and mixing it into training yields no gain.

LIMITATION

Only visual question answering is studied — extending to more tasks and modalities is future work.

Thanks — questions?

Hongyuan Lu · Shujie Miao · Wai Lam · Multi·LLL 2026